The gradient

Where linear algebra and calculus meet — and how models train.

THE IDEA

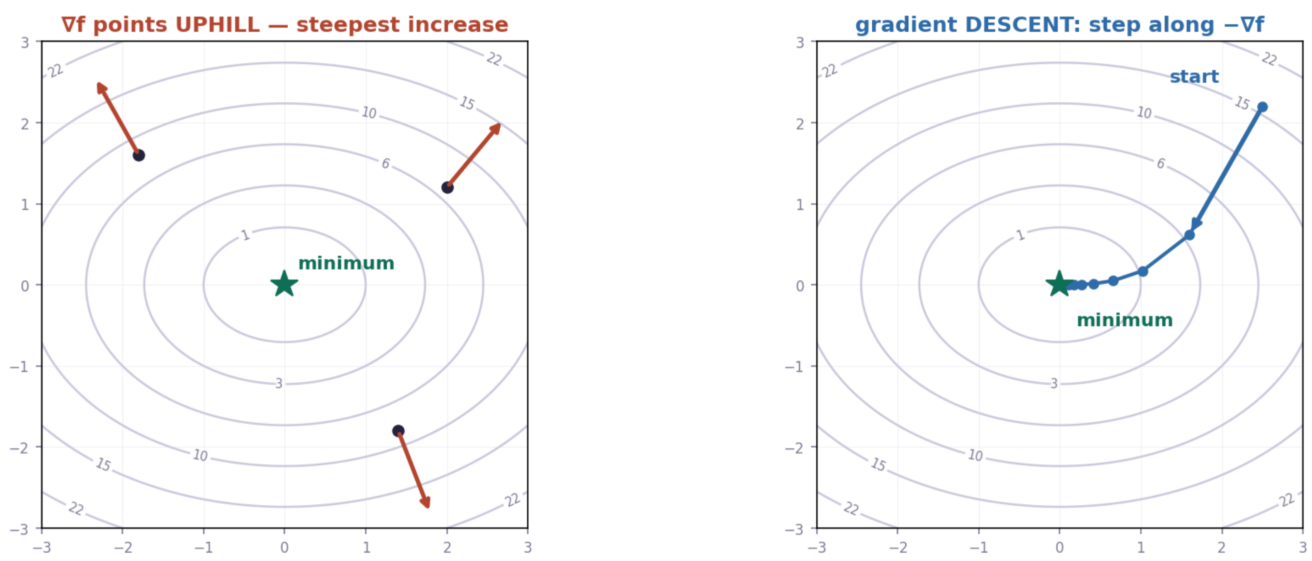
Collect all the partial derivatives into a single **vector**. That vector is the **gradient**, written ∇f (say "grad f" or "del f").

> **$\nabla f = (\partial f/\partial x , \partial f/\partial y)$** — a vector whose components are the partial derivatives.

WHAT IT TELLS YOU

Property	Meaning
Direction	Points the way of **steepest increase**
Length	How steep that increase is
$-\nabla f$	Points the way of steepest **decrease** — downhill
$\nabla f = 0$	Flat: a minimum, a maximum, or a saddle point

$f(x, y) = x^2 + 2y^2$ — the gradient and how models train



EVALUATING A GRADIENT

The gradient is a formula until you plug in a point. Substituting a specific (x, y) turns it into an ordinary vector of numbers — exactly the kind you handled in Linear Algebra.

WORKED EXAMPLE

THE GRADIENT — ∇f

where linear algebra and calculus meet

THE DEFINITION

$$\nabla f = \left(\partial f/\partial x \,,\ \partial f/\partial y \right)$$

a VECTOR whose components are the partial derivatives

WHAT IT MEANS

direction	points the way of STEEPEST INCREASE
length	how steep that increase is
negative $-\nabla f$	points the way of steepest DECREASE

EXAMPLE $f(x, y) = x^2y + 3y^2$

from the previous page :

$\partial f/\partial x = 2xy$ $\quad$ $\partial f/\partial y = x^2 + 6y$

$$\nabla f = \left(2xy \,,\ x^2 + 6y \right)$$

EVALUATE AT A POINT $(x, y) = (1, 2)$

$\partial f/\partial x = 2(1)(2) = 4$

$\partial f/\partial y = (1)^2 + 6(2) = 13$

$$\nabla f\,(1,2) = \left(4 \,,\ 13 \right)$$

a plain vector — exactly like week one

GRADIENT DESCENT : $w\_new = w\_old - learning\_rate \cdot \nabla f$

an anchor point, a direction, and a dial — the parametric line from Linear Algebra chapter 2. This is how models train.

Gradient descent — how every model trains: w_new = w_old – learning_rate · ∇f
An anchor point, a direction, and a dial. That is the parametric line from Linear Algebra Chapter 2, run in a space of millions of dimensions. The minus sign is what makes it descend rather than climb, and the learning rate is how big each step is.

KEY POINTS FROM THIS TOPIC

Gradient	$\nabla f = (\partial f/\partial x \,, \partial f/\partial y)$
Direction	steepest increase
$-\nabla f$	steepest decrease — downhill
$\nabla f = 0$	flat point: minimum, maximum or saddle
Gradient descent	w_new = w_old – learning_rate · ∇f

TEST — Topic 2

1. $f(x, y) = x^2 + y^2$. Find ∇f. → **(2x, 2y)**
2. $f(x, y) = x^2y + 5y$. Find ∇f, then evaluate it at (2, 1). → **$\nabla f = (2xy, x^2 + 5)$; at (2,1) = (4, 9)**
3. $f(x, y) = 3x^2y$. Find ∇f and evaluate at (1, 2). → **$\nabla f = (6xy, 3x^2)$; at (1,2) = (12, 3)**